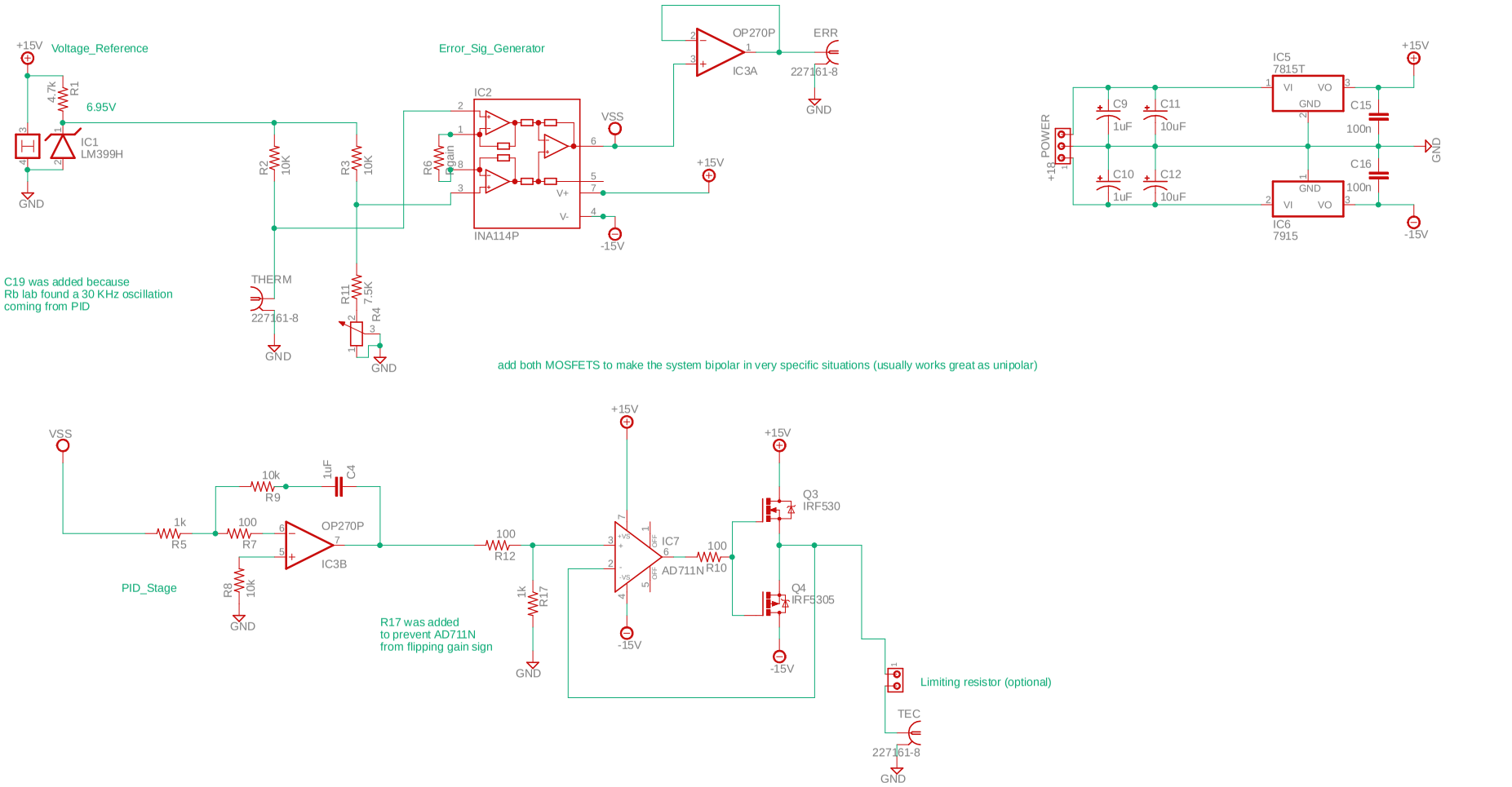
Voltage_Reference
+15V
4.7k
R1
6.95V
IC1
LM399H
GND
Error_Sig_Generator
IC2
R2
10K
R3
10K
R6
Rgain
INA114P
VSS
V+
V-
+15V
-15V
OP270P
IC3A
ERR
227161-8
GND
C19 was added because
Rb lab found a 30 KHz oscillation
coming from PID
THERM
227161-8
R11
7.5K
R4
Use Jumpers if using a stable power supply and not populate the voltage regulator
POWER
+18
C9
1uF
C11
10uF
C10
1uF
C12
10uF
IC5
7815T
VI
VO
GND
C15
100n
C16
100n
IC6
7915
add both MOSFETS to make the system bipolar in very specific situations (usually works great as unipolar)
VSS
10k
R9
1uF
C4
1k
R5
100
R7
OP270P
IC3B
R8
10k
PID_Stage
100
R12
1k
R17
R17 was added
to prevent AD711N
from flipping gain sign
IC7
AD711N
100
R10
Q3
IRF530
Q4
IRF5305
Limiting resistor (optional)
TEC
227161-8